Symmetric RNNs

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1 Reduction to Smaller Network

The symmetries we've noticed of the argmedian and argoutlier models (of sequence length 3 and width 4) are equivalent to the equations

$$\begin{aligned} w_{hh} &= U^T w'_{hh} U \\ w_{ih} &= U^T w'_{ih} \\ w_{ho} &= w'_{ho} V \end{aligned}$$

where $U = \begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & -1 & 0 \end{pmatrix}$, $V = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix}$ and $w'_{hh}, w'_{ih}, w'_{ho}$ are 2x2, 2x1, 3x2 matrices respectively. Notice that

$$\begin{aligned} U\text{ReLU}\left(U^T \underline{v}\right) &= \underline{v} \\ V\text{ReLU}\left(U^T \underline{v}\right) &= |\underline{v}| \end{aligned}$$

where by $|\underline{v}|$ I mean **coordinate-wise absolute value** rather than norm. Therefore the function defined by the RNN is

$$\begin{aligned} &w_{ho}\text{ReLU}\left(w_{ih}x_2 + w_{hh}\text{ReLU}\left(w_{ih}x_1 + w_{hh}\text{ReLU}\left(w_{ih}x_0\right)\right)\right) \\ &\quad = w'_{ho}V\text{ReLU}\left(U^T w'_{ih}x_2 + U^T w'_{hh}U\text{ReLU}\left(U^T w'_{ih}x_1 + U^T w'_{hh}U\text{ReLU}\left(U^T w'_{ih}x_0\right)\right)\right) \\ &\quad = w'_{ho}\left|\left(w'_{ih}x_2 + w'_{hh}\left(w'_{ih}x_1 + w'_{hh}\left(w'_{ih}x_0\right)\right)\right)\right| \end{aligned}$$

The linear function inside the absolute value is almost generic in the sense that there is only a codimension > 0 space of linear functions $\mathbb{R}^3 \to \mathbb{R}^2$ that are not of this form. So, RNNs of this size with the symmetries above are generically equivalent to a (3,2,3) neural network with activation function given by absolute value.

2 Explanation of AbsActivation Model for argoutlier

This is a feedforward neural network with input and output size 3 and hidden size 2 (with a single hidden layer). When trained on argoutlier this gets 99.5%

accuracy at least. I want to explain its weights. The first matrix is 2x3 and is called fc1 and the second is 3x2 and called fc2.

A few things to notice first: all neural networks has the in-out symmetry that you can balance by a positive scalar on the inputs and outputs of a single hidden neuron. Using this we can assume that fc1[0,0]=1 and fc1[1,1]=1 (the reason for not just picking the first element of each row will become clear). This reduces us to 10 parameters. Also, if a constant multiple of a hidden neuron is added to all output neurons that doesn't affect the accuracy of the model, and so we can translate the columns of fc2 to have sum zero. This reduces us to 8 parameters. When we do this, we see that the trained model becomes

$$fc1 = \begin{pmatrix} 1 & \lambda & -(1+\lambda) \\ \lambda & 1 & -(1+\lambda) \end{pmatrix}, \qquad fc2 = \begin{pmatrix} -\mu & \mu - \epsilon \\ \mu - \epsilon & -\mu \\ \epsilon & \epsilon \end{pmatrix}.$$

In the trained model, $\lambda \approx 1.44, \mu \approx 69.72$ and $\epsilon \approx 2.72$, however in what follows we will use only the fact that they are all positive, and that $\lambda > 1$.

2.1 Quadrants for the Hidden Layers

First let's look at the pre-activations for the hidden neurons based on their signs.

2.1.1 Positive Quadrant

In the quadrant where the pre-activation of both hidden neurons are positive, we have the following:

$$\begin{aligned} x_0 + \lambda x_1 - (1+\lambda)x_2 > 0, \\ \lambda x_0 + x_1 - (1+\lambda)x_2 > 0. \end{aligned}$$

This implies in particular that $x_2 < \overline{x}$. In fact it implies something slightly stronger, that

$$x_2 < \overline{x} - \frac{\lambda - 1}{3(1+\lambda)}|x_0 - x_1|.$$

In fact it is equivalent to this statement. Notice that with our specific values, the coefficient of $|x_0 - x_1|$ is approximately 0.06.

2.1.2 $(+,-)$-Quadrant

In the quadrant where

$$\begin{aligned} x_0 + \lambda x_1 - (1+\lambda)x_2 > 0, \\ \lambda x_0 + x_1 - (1+\lambda)x_2 < 0, \end{aligned}$$

we deduce that $x_1 > x_2 > x_0$. In fact these inequalities are equivalent to the fact that

$$x_1 > x_0, \qquad \text{and} \qquad |x_2 - \overline{x}| < \frac{\lambda - 1}{3(1+\lambda)}(x_1 - x_0)$$

2.1.3 $(-,+)$-Quadrant

This is exactly the same as the previous with x_0, x_1 switched.

2.1.4 Negative Quadrant

This is analogous to the positive quadrant case, the inequalities are equivalent to

$$x_2 > \overline{x} + \frac{\lambda - 1}{3(1+\lambda)}|x_0 - x_1|.$$

2.2 Logits

2.2.1 Positive Quadrant

Notice that on this quadrant (where $x_2 < \overline{x}$) we see that the argoutlier is decided in cases: if $x_2 < \min(x_0, x_1) - |x_0 - x_1|$ then x_2 is the outlier, and otherwise the outlier is $\max(x_0, x_1)$. In other words, the first logit should be highest if and only if $x_0 - x_1 > \overline{x} - x_2$, the second logit should be highest if and only if $x_1 - x_0 > \overline{x} - x_2$ and the final logit should be highest if and only if $|x_1 - x_0| < \overline{x} - x_2$.

In this quadrant, the actual logits we get are

$$\text{logits} = \begin{pmatrix} (-\mu + \mu\lambda - \epsilon\lambda)x_0 + (-\mu\lambda + \mu - \epsilon)x_1 + \epsilon(1+\lambda)x_2 \\ (-\mu\lambda + \mu - \epsilon)x_0 + (-\mu + \mu\lambda - \epsilon\lambda)x_1 + \epsilon(1+\lambda)x_2 \\ \epsilon(1+\lambda)(x_0 + x_1 - 2x_2) \end{pmatrix}$$

The final logit is always positive and the logits sum to zero, so one of the first two logits is highest iff it is higher than the final logit, and otherwise the final logit is highest. Let us consider the difference between the first logit and the final one. This is

$$\begin{aligned} &(-\mu + \mu\lambda - 2\epsilon\lambda - \epsilon)x_0 + (-\mu\lambda + \mu - 2\epsilon - \epsilon\lambda)x_1 + 3\epsilon(1+l)x_2 \\ &\qquad = \mu(\lambda - 1)(x_0 - x_1) + \frac{1}{2}\epsilon(1-\lambda)(x_0 - x_1) - \frac{3}{2}\epsilon(1+\lambda)(x_0 + x_1 - 2x_2) \\ &\qquad = \left(\mu - \frac{1}{2}\epsilon\right)(\lambda - 1)(x_0 - x_1) - \frac{9}{2}\epsilon(1+\lambda)(\overline{x} - x_2) \end{aligned}$$

Therefore the first logit is highest if and only if

$$x_0 - x_1 > \frac{9\epsilon(1+\lambda)}{(2\mu - \epsilon)(\lambda - 1)}(\overline{x} - x_2).$$

For this the specific quantities of our model, this ratio is almost exactly 1! Similarly the second logit is highest iff

$$x_1 - x_0 > \frac{9\epsilon(1+\lambda)}{(2\mu - \epsilon)(\lambda - 1)}(\overline{x} - x_2).$$

Consequently the third logit is highest iff

$$|x_1 - x_0| < \frac{9\epsilon(1+\lambda)}{(2\mu-\epsilon)(\lambda-1)}(\overline{x} - x_2)$$

Since the coefficient here is almost exactly 1 in the trained model, the accuracy is perfect on this region (and the opposite region by symmetry).

2.2.2 $(+,-)$-Quadrant

In this quadrant it suffices to know whether $x_2 > \overline{x}$ or vice-versa. In this region the logits are as follows:

$$\text{logits} = \begin{pmatrix} (-\mu - \lambda\mu + \lambda\epsilon)x_0 + (-\mu\lambda - \mu + \epsilon)x_1 + (1+\lambda)(2\mu-\epsilon)x_2 \\ (\mu - \epsilon + \mu\lambda)x_0 + (\mu + \lambda\mu - \lambda\epsilon)x_1 - (1+\lambda)(2\mu-\epsilon)x_2 \\ (\lambda-1)\epsilon(x_1 - x_0) \end{pmatrix}$$

If $\epsilon = 0$, we get that the final entry is 0, and that the first entry is

$$\mu(1+\lambda)(-x_0 - x_1 + 2x_2) = 3\mu(1+\lambda)(x_2 - \overline{x}),$$

and so the result would be perfectly accurate. In fact, the first entry is

$$\begin{aligned} &\mu(1+\lambda)(-x_0 - x_1 + 2x_2) + \epsilon(\lambda x_0 + x_1 - (1+\lambda)x_2) \\ &\quad = 3\mu(1+\lambda)(x_2 - \overline{x}) + \epsilon(\lambda x_0 + x_1 - (1+\lambda)x_2) \end{aligned}$$

This means that the difference between the first and third entry would be

$$\begin{aligned} &3\mu(1+\lambda)(x_2 - \overline{x}) + \epsilon((2\lambda-1)x_0 + (2-\lambda)x_1 - (1+\lambda)x_2) \\ &\quad = \left(3\mu - \frac{1}{2}\epsilon\right)(1+\lambda)(x_2 - \overline{x}) - \frac{3}{2}\epsilon(\lambda-1)(x_1 - x_0) \end{aligned}$$

Similarly the difference between the second and third entry would be

$$-\left(3\mu - \frac{1}{2}\epsilon\right)(1+\lambda)(x_2 - \overline{x}) - \frac{3}{2}\epsilon(\lambda-1)(x_1 - x_0)$$

The result of this is that the first logit is highest precisely when

$$x_2 - \overline{x} > \frac{3\epsilon(\lambda-1)}{(6\mu-\epsilon)(1+\lambda)}(x_1 - x_0).$$

The second logit is highest precisely when

$$x_2 - \overline{x} < -\frac{3\epsilon(\lambda-1)}{(6\mu-\epsilon)(1+\lambda)}(x_1 - x_0)$$

and the third logit is highest when

$$|x_2 - \overline{x}| < \frac{3\epsilon(\lambda-1)}{(6\mu-\epsilon)(1+\lambda)}(x_1 - x_0)$$

This interval of failure is 0.3% of the range. The smaller this coefficient becomes the better.

2.2.3 $(-,+)$-Quadrant

This is exactly analogous to the case above with x_0, x_1 switched.

2.2.4 Negative Quadrant

This is exactly the same as the positive quadrant!

2.3 Extras

Suppose that we impose that the coefficient in the positive quadrant logits is 1, so that the model has perfect accuracy in that region. Then

$$9\epsilon(1+\lambda) = (2\mu - \epsilon)(\lambda - 1) \implies \epsilon = \mu\frac{\lambda - 1}{5\lambda + 4}.$$

Then, the coefficient in the $(+,-)$-quadrant becomes

$$\frac{3(\lambda-1)^2(5\lambda+4)}{(6(5\lambda+4)-(\lambda-1))(1+\lambda)} = \frac{3(\lambda-1)^2(5\lambda+4)}{(29\lambda+25)(1+\lambda)}$$

This can be driven down closer to zero by moving λ towards 1.

3 Explanation of AbsActivation Model for argmedian

Here the form of the solution after doing the same sort of normalising as in the argoutlier case is

$$fc1 = \begin{pmatrix} 1 & -(\lambda+1) & \lambda \\ 1 & \lambda & -(\lambda+1) \end{pmatrix}, \qquad fc2 = \begin{pmatrix} \epsilon & \epsilon \\ -\mu & \mu - \epsilon \\ \mu - \epsilon & -\mu \end{pmatrix}.$$

The specific values we get are $\lambda \approx 7.808, \epsilon \approx 0.7$, and $\mu \approx 17.79$.

3.1 Quadrants for the Hidden Layer

3.1.1 Positive Quadrant

If both preactivations are positive, then we get

$$x_0 - (\lambda+1)x_1 + \lambda x_2 > 0$$
$$x_0 + \lambda x_1 - (\lambda+1)x_2 > 0$$

This is equivalent to the statement that

$$x_0 > \frac{x_1 + x_2}{2} + \frac{2\lambda+1}{2}|x_1 - x_2|.$$

With our values the coefficient here is approximately 8.308.

3.1.2 $(+,-)$-Quadrant

Here, the inequalities are

$$x_0 - (\lambda+1)x_1 + \lambda x_2 > 0$$
$$x_0 + \lambda x_1 - (\lambda+1)x_2 < 0$$

From this we deduce that $x_2 > x_1$ and

$$\left| x_0 - \frac{x_1+x_2}{2} \right| < \frac{2\lambda+1}{2}|x_1 - x_2|.$$

3.1.3 Other Quadrants

Are exactly analogous.

3.2 Logits

3.2.1 Positive Quadrant

On the positive quadrant, we see that $x_0 > x_1, x_2$ and so the median is simply the larger of x_1 or x_2.

On this region, the logits are given by

$$\begin{pmatrix} \epsilon(2x_0 - x_1 - x_2) \\ -\epsilon x_0 + (2\mu\lambda + \mu - \lambda\epsilon)x_1 + (-2\lambda\mu - \mu + \lambda\epsilon + \epsilon)x_2 \\ -\epsilon x_0 + (-2\mu\lambda - \mu + \lambda\epsilon + \epsilon)x_1 + (2\lambda\mu - \lambda\epsilon + \mu)x_2 \end{pmatrix}$$

The first logit is positive, and they sum to zero, so one of the second logits is largest of all precisely when it is larger than the first. The difference between the second logit and the first is

$$\begin{aligned} -3\epsilon x_0 &+ (2\mu\lambda + \mu - \lambda\epsilon + \epsilon)x_1 + (-2\lambda\mu - \mu + \lambda\epsilon + 2\epsilon)x_2 \\ &= 3\epsilon\left(-x_0 + \frac{1}{2}x_1 + \frac{1}{2}x_2\right) + (2\mu\lambda + \mu - \lambda\epsilon - \frac{1}{2}\epsilon)(x_1 - x_2) \\ &= 3\epsilon\left(\frac{x_1+x_2}{2} - x_0\right) + \frac{1}{2}(2\lambda+1)(2\mu-\epsilon)(x_1 - x_2) \end{aligned}$$

Therefore, on this region, the second logit is largest precisely when

$$x_1 - x_2 > \frac{6\epsilon}{(2\lambda+1)(2\mu-\epsilon)}\left(x_0 - \frac{x_1+x_2}{2}\right).$$

This ratio is 0.007. The third logit is the same, and this shows that the only interval of failure here is when

$$|x_1 - x_2| < \frac{6\epsilon}{(2\lambda+1)(2\mu-\epsilon)}\left(x_0 - \frac{x_1+x_2}{2}\right)$$

3.2.2 $(+,-)$-Quadrant

On this quadrant, the logits are given by

$$\begin{pmatrix} (2\lambda+1)\epsilon(x_2-x_1) \\ (-2\mu+\epsilon)x_0+(\mu+\lambda\epsilon)x_1+(\mu-\lambda\epsilon-\epsilon)x_2 \\ (2\mu-\epsilon)x_0+(-\mu+\lambda\epsilon+\epsilon)x_1-(\mu+\lambda\epsilon)x_2 \end{pmatrix}$$

The first logit is always positive, and the logits sum to zero, and so each of the second and third logits are maximal iff they are larger than the first. The difference between the second and first logits is

$$\begin{aligned}(-2\mu+\epsilon)x_0+(\mu+3\lambda\epsilon+\epsilon)x_1+(\mu-3\lambda\epsilon-2\epsilon)x_2 \\ = (2\mu-\epsilon)\left(\frac{x_1+x_2}{2}-x_0\right)-\frac{3}{2}\epsilon(2\lambda+1)(x_2-x_1)\end{aligned}$$

This is positive iff

$$x_2-x_1<\frac{2(2\mu-\epsilon)}{3\epsilon(2\lambda+1)}\left(\frac{x_1+x_2}{2}-x_0\right).$$

This ratio is almost exactly 2, which gives perfect accuracy. Similarly for the difference between the third and first logits, which is positive iff

$$x_1-x_2>\frac{2(2\mu-\epsilon)}{3\epsilon(2\lambda+1)}\left(\frac{x_1+x_2}{2}-x_0\right).$$

again achieving perfect accuracy.